








Statistics of Financial Markets

Seminar 7: Efficient markets, random walk and variance-ratio tests



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **BET**: Bucharest Exchange Trading (index)
- **BIST**: Borsa İstanbul (BIST 100 index) — the Istanbul Stock Exchange (BIST 100 index)
- **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CD**: Chow–Denning (multiple variance-ratio test)
- **CDF**: Cumulative Distribution Function
- **COVID-19**: COronaVIrus Disease 2019
- **EODHD**: EOD Historical Data (market-data provider)
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)

Acronyms used in this material (2/2)

- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (stationarity test)
- **LB**: Ljung–Box (test)
- **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- **PP**: Phillips–Perron (unit-root test)
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- **RW1**: Random Walk 1 (i.i.d. increments)
- **RW2**: Random Walk 2 (independent increments)
- **RW3**: Random Walk 3 (uncorrelated increments)
- **VR**: Variance Ratio
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's Question and Seminar Structure

- Question of the seminar
  - **how can we test whether past prices predict future returns without fooling ourselves?**
- Outputs to produce
  - the answers to the A exercises
  - the table or figure each B exercise asks for
  - a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Nothing is handed in and nothing is graded: the seminar is for practice
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the Exercise Map (next slide)
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	the square-root-of-time rule and variance ratios from autocorrelations	Solved, Proposed	A1
A3, A4	Box–Pierce and Ljung–Box statistics, step by step	Solved, Proposed	A3
A5, A6	the runs test on a sequence of signs	Solved, Proposed	A5
A7, A8	ADF and KPSS decisions from regression output	Solved, Proposed	A7
B1, B2	ACF, Ljung–Box and robust tests, runs test: S&P 500; BET, Bitcoin, BVB stocks	Solved, Proposed	B1
B3, B4	ADF, Phillips–Perron and KPSS on prices and returns: DAX; BET, Bitcoin, TLV	Solved, Proposed	B3
B5, B6	variance-ratio tests: S&P 500; five emerging markets	Solved, Proposed	B5
C1, C2	has the BET become more efficient? what is wrong in an AI answer?	Proposed	B1, B5

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations on paper; Part B: real data in the notebook, each task ending with an interpretation question; Part C: open questions and the audit of an AI answer

Data Used

Series	Source	Price used	Period
S&P 500, DAX	EODHD	close	1990–2026
BET	EODHD	close	1997–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026
Banca Transilvania (TLV), OMV Petrom (SNP)	EODHD	adjusted close	2010–2026
WIG20, BUX, PX, BIST 100	EODHD	close	last ten years

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped (except Bitcoin); the last day is 18 September 2026
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, Chapter 2)
- In the notebook: `returns('sp500')`, `log_price('dax')`; no account or key is needed

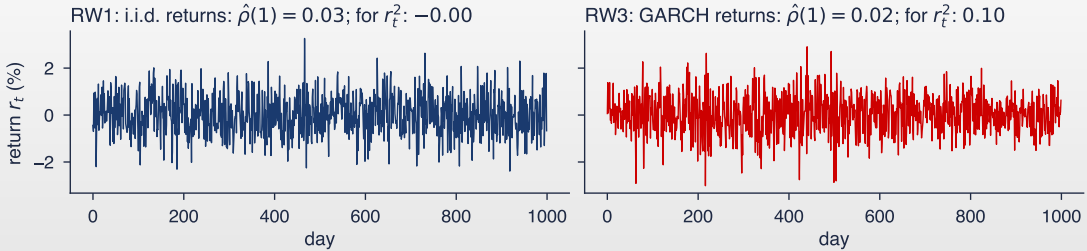
Prerequisites for Today (1/18): Weak-Form Efficiency

- **Weak-form efficiency** Fama (1970): past prices cannot be used to earn returns above the return that pays for risk
  - r_{t+1}: tomorrow's log return; μ: its equilibrium expected value; $r_{t+1} - \mu$: the excess return
  - excess returns are a **martingale difference**: $E[r_{t+1} - \mu \mid \text{past}] = 0$
  - in words: no information in past prices predicts the excess return
- The variance may still be predictable: volatility clustering is compatible with efficiency

Prerequisites for Today (2/18): Random Walks

- Log price $p_t = \ln P_t$, **random walk**: $p_t = \mu + p_{t-\mathbf{1}} + \varepsilon_t$, so $r_t = \mu + \varepsilon_t$ Campbell, Lo & MacKinlay (1997)
  - μ: the drift (mean daily return); ε_t: the shock of day t, mean 0, variance $\sigma^{\mathbf{2}}$
- Three versions, from strong to weak
  - **RW1**: ε_t i.i.d. (independent, identically distributed); **RW2**: independent; **RW3**: uncorrelated
  - RW1 $\Rightarrow$ RW2 $\Rightarrow$ RW3; volatility clustering is allowed only by RW3 and by the martingale
- Under any of them, $\mathrm{Var}(r_t(q)) = q\sigma^{\mathbf{2}}$, with $r_t(q) = r_t + \cdots + r_{t-q+\mathbf{1}}$ the q-day return
  - "square-root-of-time rule": $\mathrm{sd}(r_t(q)) = \sigma\sqrt{q}$

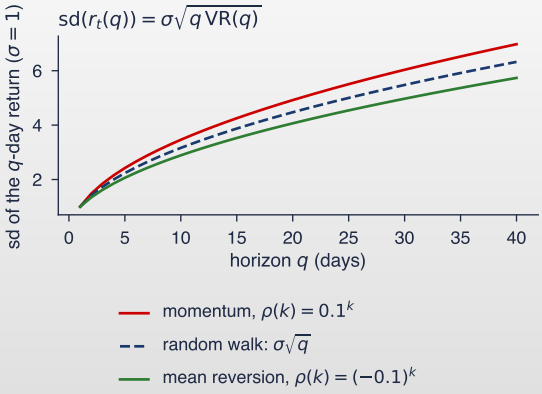
Prerequisites for Today (3/18): RW1 and RW3 on a Chart



- Left: i.i.d. Normal returns (RW1); right: GARCH(1,1) returns, uncorrelated but with volatility clustering (RW3, not RW1)
- GARCH(1,1): $r_t = \sigma_t \varepsilon_t$, $\sigma_t^2 = \omega + a\, r_{t-1}^2 + b\, \sigma_{t-1}^2$, ε_t i.i.d. $N(0,1)$
  - here $\omega = 0.02$, $a = 0.08$, $b = 0.9$: a large return raises the variance of the next day
  - both series have $\hat{\rho}(1) \approx 0$; only the squared returns tell them apart

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Prerequisites for Today (4/18): the Square-Root-of-Time Rule



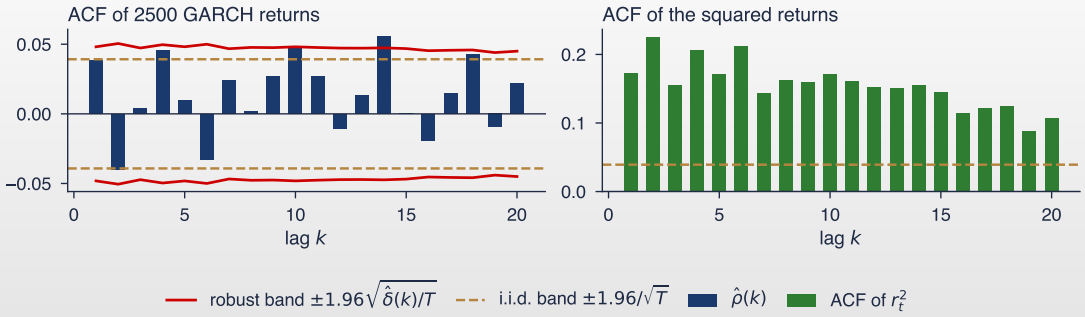
- With autocorrelated returns, the q-day return $r_t(q)$ of 2/18 has
$$\mathrm{Var}(r_t(q)) = q\sigma^2\,\mathrm{VR}(q)$$
  - VR(q): the variance ratio (15/18); random walk: $\mathrm{VR}(q) = 1$, the dashed line $\sigma\sqrt{q}$
- Positive autocorrelations (red): the q-day risk is larger than $\sigma\sqrt{q}$
- Negative autocorrelations (green): it is smaller
- Chart: returns with $\rho(k) = \phi^k$, $\phi = 0.1$ and $\phi = -0.1$

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Prerequisites for Today (5/18): the Sample ACF

- Sample **ACF** (autocorrelation function): $\hat{\rho}(k) = \sum_{t>k} e_t e_{t-k} / \sum_t e_t^2$
  - $e_t = r_t - \bar{r}$: the demeaned return; k: the lag in days; T: the number of returns
  - $\hat{\rho}(k)$: the correlation between today's return and the return k days earlier
- i.i.d. returns: $\mathrm{Var}(\hat{\rho}(k)) \approx 1/T$
  - 95% band $\pm 1.96/\sqrt{T}$: values outside it are significant at 5%
- Robust (volatility clustering): $\mathrm{Var}(\hat{\rho}(k)) \approx \hat{\delta}(k)/T$ Lo & MacKinlay (1988)
  - $\hat{\delta}(k) = T \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$: about 1 for i.i.d. returns, above 1 when large returns follow large returns

Prerequisites for Today (6/18): i.i.d. and Robust Bands on a Chart



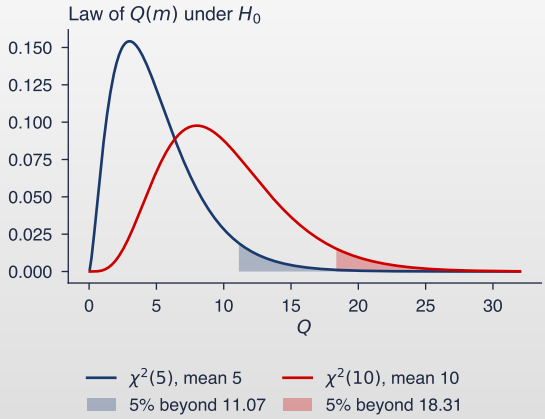
- Left: the ACF of 2 500 simulated GARCH returns, uncorrelated by construction
  - the i.i.d. band (dashed) is too narrow: some $\hat{\rho}(k)$ fall outside it by chance
  - the robust band (red) is wider: here $\hat{\delta}(1) = 1.51$
- Right: the squared returns are strongly autocorrelated: volatility clustering, which makes $\hat{\delta}(k) > 1$

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Prerequisites for Today (7/18): Portmanteau Tests

- H_0: $\rho(1) = \cdots = \rho(m) = 0$: no autocorrelation in the first m lags
  - three statistics, each $\approx \chi^2(m)$ (chi-square with m degrees of freedom) under its null
- Box–Pierce $Q_{BP} = T \sum_{k=1}^{m} \hat{\rho}(k)^2$ Box & Pierce (1970)
  - **LB** (Ljung–Box) $Q_{LB} = T(T+2) \sum_{k=1}^{m} \hat{\rho}(k)^2/(T-k)$ Ljung & Box (1978): the same idea, closer to χ^2 in small samples
- Robust $\tilde{Q} = \sum_{k=1}^{m} T\hat{\rho}(k)^2/\hat{\delta}(k)$ Escanciano & Lobato (2009)
  - each squared autocorrelation divided by its own variance: valid for a martingale with volatility clustering
- 5% critical values: $\chi^2_{0.95}(3) = 7.81$, $\chi^2_{0.95}(5) = 11.07$, $\chi^2_{0.95}(10) = 18.31$

Prerequisites for Today (8/18): the Chi-Square Law of Q



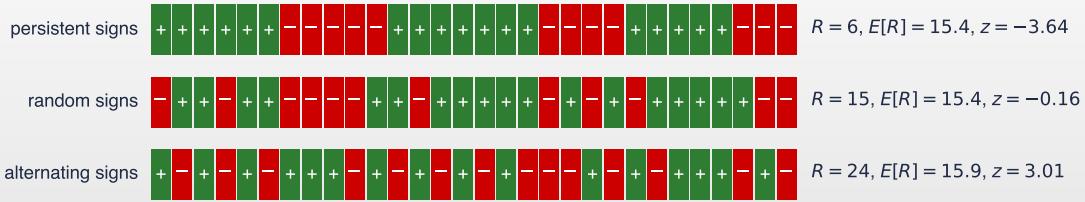
- Why $\chi^2(m)$: under H_0 each $T\hat{\rho}(k)^2$ is roughly a squared $N(0,1)$ variable
  - their sum over m lags is $\chi^2(m)$, with mean m
- Shaded: the 5% tails beyond the critical values of the previous slide
- p-value: the area to the right of the observed Q
- A single large $\hat{\rho}(k)$ can be enough to push Q into the tail

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Prerequisites for Today (9/18): the Runs Test

- A **run**: a maximal sequence of returns with the same sign Wald & Wolfowitz (1940); zero returns are dropped
  - $++---+-$: $R = 4$ runs; $n_1 = 3$ positive, $n_2 = 4$ negative, $n = n_1 + n_2 = 7$
- Under independence of the signs:
  - $E[R] = \dfrac{2n_1 n_2}{n} + 1, \quad \mathrm{Var}(R) = \dfrac{2n_1 n_2(2n_1 n_2 - n)}{n^2(n-1)}$
  - $z = (R - E[R])/\sqrt{\mathrm{Var}(R)} \approx N(0,1)$; reject at 5% if $|z| > 1.96$
- Reading z
  - $z < 0$: too few runs, signs persist (positive dependence, trends)
  - $z > 0$: too many runs, signs alternate (negative dependence, reversals)
- Signs only: the test ignores the size of the returns, so heavy tails and volatility do not affect it

Prerequisites for Today (10/18): Runs on a Chart



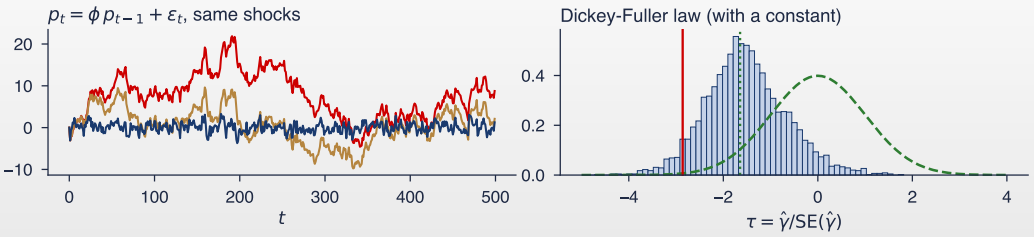
- Three sequences of 30 signs (green $+$, red $-$); R counts the blocks of equal signs
- Persistent signs: few long runs; alternating signs: many short runs; both reject independence, in the two directions of the previous slide
- Random signs: R close to $E[R]$, $|z| < 1.96$: independence is not rejected

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Prerequisites for Today (11/18): Unit Roots

- $I(0)$: stationary (constant mean and variance, shocks fade)
  - $I(1)$: the first difference $\Delta p_t = p_t - p_{t-\mathbf{1}}$ is $I(0)$; a random walk is $I(1)$
- Dickey–Fuller regression Dickey & Fuller (1979): $\Delta p_t = c\ [+\, bt] + \gamma\, p_{t-\mathbf{1}}\ [+ \sum_j \delta_j \Delta p_{t-j}] + \varepsilon_t$
  - c: constant; bt: optional trend; δ_j: coefficients of the optional lagged differences
  - $\gamma = 0$: a unit root (shocks are permanent); $\gamma < 0$: the series returns to a level
  - $H_{\mathbf{0}}$: $\gamma = 0$; $\tau = \hat{\gamma}/\mathrm{SE}(\hat{\gamma})$; reject if τ is below the critical value
- 5% critical values MacKinnon (1996): -2.86 (constant), -3.41 (constant and trend); not -1.645

Prerequisites for Today (12/18): Unit Roots on a Chart



- Left: the same shocks with $\phi = 1$ (a random walk: shocks never fade), $\phi = 0.98$ (they fade slowly) and $\phi = 0.5$ (fast)
  - in the Dickey–Fuller regression $\gamma = \phi - 1$: $\gamma = 0$ is the unit root
- Right: 4 000 simulations of τ under H_0: the law lies to the left of $N(0,1)$
  - simulated 5% quantile -2.84, close to -2.86; with -1.645 a true unit root would be rejected far more often than 5%

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Prerequisites for Today (13/18): ADF, PP and KPSS

- **ADF** (augmented Dickey–Fuller): the regression with the lagged differences Said & Dickey (1984)
  - **PP** (Phillips–Perron): without them, with a corrected τ Phillips & Perron (1988); same H_0 and critical values
- **KPSS** Kwiatkowski et al. (1992): the hypotheses are reversed, H_0 stationarity
  - reject if the statistic exceeds 0.463 (constant) or 0.146 (trend)
- Combining the two
  - ADF does not reject + KPSS rejects: $I(1)$; ADF rejects + KPSS does not: $I(0)$; otherwise: inconclusive

Prerequisites for Today (14/18): the KPSS Statistic and the Lag Choice

- **KPSS**: e_t = residuals of p_t on a constant (and a trend)

$$\mathrm{KPSS} = \frac{1}{T^2 \hat{\sigma}^2_{LR}} \sum_{t=1}^{T} S_t^2, \qquad S_t = e_1 + \cdots + e_t$$

  - S_t: the partial sums; with a unit root they wander far from 0, so KPSS is large

- Long-run variance with l Bartlett lags (Newey–West)

$$\hat{\sigma}^2_{LR} = \hat{\gamma}_0 + 2 \sum_{j=1}^{l} \left(1 - \frac{j}{l+1}\right) \hat{\gamma}_j$$

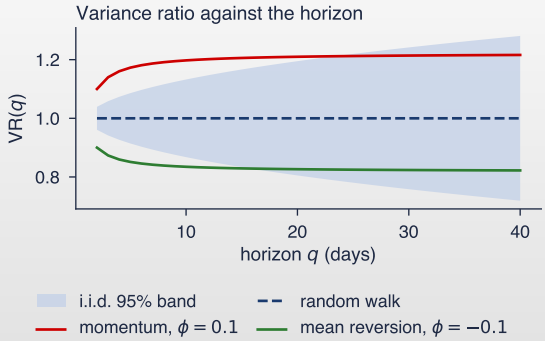
  - $\hat{\gamma}_j = \frac{1}{T} \sum_{t>j} e_t e_{t-j}$: the autocovariance at lag j; PP corrects τ with the same kind of estimate

- ADF lag order by AIC (Akaike information criterion)
  - $\mathrm{AIC} = T \ln \hat{\sigma}^2_\varepsilon + 2k$ for each candidate number of lags; keep the lowest
  - $\hat{\sigma}^2_\varepsilon$: the residual variance; k: the number of estimated coefficients

Prerequisites for Today (15/18): Variance Ratios

- **VR** (variance ratio): $\mathrm{VR}(q) = \dfrac{\mathrm{Var}(r_t(q))}{q\,\mathrm{Var}(r_t)} = 1 + 2\sum_{k=1}^{q-1}\left(1 - \dfrac{k}{q}\right)\rho(k)$ Lo & MacKinlay (1988)
  - the q-day variance relative to q daily variances; $= 1$: random walk; > 1: momentum; < 1: mean reversion
- Tests with T returns, both $\approx N(0,1)$ under H_0
  - i.i.d.: $Z(q) = (\widehat{\mathrm{VR}}(q) - 1)/\sqrt{2(2q-1)(q-1)/(3qT)}$
  - robust: $Z^*(q) = (\widehat{\mathrm{VR}}(q) - 1)/\sqrt{\hat\theta(q)/T}$, $\hat\theta(q) = \sum_{j=1}^{q-1}[2(q-j)/q]^2\hat\delta(j)$
  - $\widehat{\mathrm{VR}}(q)$: the estimate from the data; $\hat\delta(j)$: as in the robust ACF band (5/18)

Prerequisites for Today (16/18): Variance Ratios on a Chart



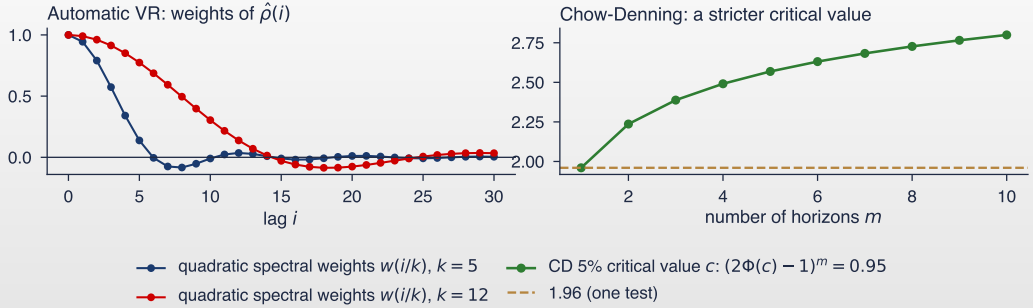
- VR(q) for returns with $\rho(k) = \phi^k$
  - $\phi = 0.1$ (red) and $\phi = -0.1$ (green): the momentum and the mean-reversion cases of the previous slide
- Shaded: the i.i.d. 95% band
  $1 \pm 1.96\sqrt{2(2q-1)(q-1)/(3qT)}$, $T = 2\,500$
  - it widens with q: long horizons are measured less precisely
  - here $\phi = \pm 0.1$ is detected at short horizons, but not at long ones

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Prerequisites for Today (17/18): Several Horizons and Several Markets

- Several horizons: **CD** (Chow–Denning) $= \max_i |Z^*(q_i)|$, $i = 1, \ldots, m$ Chow & Denning (1993)
  - critical value c from $P(\mathrm{CD} \le c) = (2\Phi(c) - 1)^m$, with Φ the $N(0,1)$ CDF; 5%, $m = 4$: $c = 2.49$
- Several markets (Šidák): each of m tests at level $1 - 0.95^{1/m}$
  - the probability of at least one false rejection then stays at 5%
- Automatic VR Choi (1999): $\widehat{\mathrm{VR}}(k) = 1 + 2\sum_{i \ge 1} w(i/k)\,\hat{\rho}(i)$
  - w: weights that decrease smoothly; the bandwidth k is chosen from the data
  - p-value by wild bootstrap Kim (2009): $r_t^* = \eta_t (r_t - \bar{r})$, $\eta_t \sim N(0,1)$, which keeps the volatility and removes any autocorrelation

Prerequisites for Today (18/18): Automatic VR and Chow–Denning on a Chart



- Left: the automatic VR weights each $\hat{\rho}(i)$ by $w(i/k)$: close to 1 for small lags, near 0 beyond about $1.2k$
  - quadratic spectral kernel; a larger bandwidth k uses more lags
- Right: the Chow–Denning critical value c for m horizons
  - $m = 1$: 1.96; $m = 4$: 2.49; it grows with m so that all m tests together keep a 5% error

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A1: Variance Ratios from Autocorrelations [Solved]

Task

- Daily returns have a standard deviation $\sigma = 1.2\%$.
- 1. Under RW1, compute the standard deviation of the 5-day and of the 20-day return.
- 2. Suppose $\rho(1) = 0.10$, $\rho(2) = 0.05$ and $\rho(k) = 0$ for $k \geq 3$; compute VR(2) and VR(5).
- 3. Treat these as estimates from $T = 2500$ i.i.d. returns and compute $Z(2)$ and $Z(5)$.
- 4. Compute the 5-day standard deviation implied by VR(5).
- Report: two standard deviations, two VR, two Z and one sentence.

Solution

- 1. $1.2\sqrt{5} = 2.68\%$; $1.2\sqrt{20} = 5.37\%$
- 2. $\mathrm{VR}(2) = 1 + 0.10 = 1.10$; $\mathrm{VR}(5) = 1 + 2(0.8 \times 0.10 + 0.6 \times 0.05) = 1.220$
- 3. $\sqrt{2 \cdot 3 \cdot 1/(6 \cdot 2500)} = 0.020$, $Z(2) = 5.00$; $\sqrt{2 \cdot 9 \cdot 4/(15 \cdot 2500)} = 0.0438$, $Z(5) = 5.02$
- 4. $1.2\sqrt{5 \times 1.220} = 2.96\%$ instead of 2.68%: momentum makes weekly risk larger than the $\sqrt{q}$ rule says.

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A2: Mean Reversion in a Volatile Asset [Proposed]

Solution discussed in class

Task

- Daily returns: $\sigma = 2\%$, $\rho(1) = -0.08$, $\rho(2) = 0.03$, $\rho(k) = 0$ for $k \geq 3$, $T = 1000$. Model: A1.
- 1. Compute VR(2) and VR(5).
- 2. Compute $Z(2)$ and $Z(5)$ and decide at 5%.
- 3. Compare the 5-day standard deviation implied by VR(5) with $2\sqrt{5}$.
- Report: two VR, two Z, two standard deviations and one sentence.

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A3: Box–Pierce and Ljung–Box, Step by Step [Solved]

Task

- $T = 500$ daily returns; $\hat\rho(1), \ldots, \hat\rho(5) = 0.12, -0.05, 0.04, 0.03, -0.06$.
- 1. Compute $Q_{BP}(5)$.
- 2. Compute $Q_{LB}(5)$.
- 3. Compare with $\chi^2_{0.95}(5) = 11.07$ and decide.
- 4. Say which lag drives the result.
- Report: two statistics, a decision and one sentence.

Solution

- 1. $T\hat\rho(k)^2$: 7.20, 1.25, 0.80, 0.45, 1.80; $Q_{BP}(5) = 11.50$
- 2. $500 \cdot 502\,\hat\rho(k)^2/(500-k)$: 7.24, 1.26, 0.81, 0.46, 1.83; $Q_{LB}(5) = 11.59$
- 3. $11.59 > 11.07$: reject H_0 at 5% ($p = 0.041$)
- 4. Lag 1 alone gives 7.20 of 11.50: one autocorrelation of 0.12 drives the rejection.

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A4: Ljung–Box on Squared Returns [Proposed]

Solution discussed in class

Task

- $T = 750$; the ACF of the **squared** returns: $0.25, 0.20, 0.18, 0.15, 0.12$ at lags 1–5. Model: A3.
- 1. Compute $Q_{LB}(5)$ and decide at 5%.
- 2. Say which of RW1, RW3 and the martingale this rejects.
- 3. Say whether weak-form efficiency is rejected.
- Report: one statistic and two sentences.

Back Exercise list

A5: The Runs Test on 20 Days [Solved]

Task

- Signs of 20 daily returns:
  $+\,+\,-\,+\,+\,+\,-\,-\,+\,-\,-\,-\,+\,+\,-\,+\,-\,-\,++.$
- 1. Count n_1, n_2 and the number of runs R.
- 2. Compute $E[R]$ and $\mathrm{Var}(R)$.
- 3. Compute z and decide at 5%.
- Report: R, $E[R]$, z and one sentence.

Solution

- 1. $n_1 = 11$, $n_2 = 9$, $R = 11$
- 2. $E[R] = 2 \cdot 11 \cdot 9/20 + 1 = 10.90$;
  $\mathrm{Var}(R) = 198 \cdot 178/(400 \cdot 19) = 4.64$
- 3. $z = (11 - 10.90)/2.15 = 0.05$ ($\mathrm{p} = 0.963$): do not reject independence.
- With 20 days the test has little power: only large departures can be detected.

Back Exercise list

A6: A Sequence with Long Runs [Proposed]

Solution discussed in class

Task

- Signs of 24 daily returns: $++-\,--++\,+--\,++\,+--\,---++--$. Model: A5.
- 1. Count n_1, n_2 and R.
- 2. Compute $E[R]$, $\mathrm{Var}(R)$ and z.
- 3. Decide at 5% and say what kind of dependence the signs show.
- Report: R, $E[R]$, z and one sentence.

A7: ADF and KPSS from Regression Output [Solved]

Task

- Log price, $T = 2500$: $\Delta p_t = 0.021 - 0.0028\, p_{t-\mathbf{1}} + 0.05\, \Delta p_{t-\mathbf{1}} + e_t$, $\mathrm{SE}(\hat{\gamma}) = 0.0019$; KPSS (constant) $= 2.10$.
- Returns: $\Delta r_t = 0.03 - 0.92\, r_{t-\mathbf{1}} + e_t$, $\mathrm{SE}(\hat{\gamma}) = 0.020$; KPSS (constant) $= 0.09$.
- 1. Compute τ for the price and for the returns.
- 2. Decide with the 5% critical value for a constant.
- 3. Decide with KPSS and give the order of integration of each series.
- Report: two τ, four decisions and one sentence.

Solution

- 1. Price: $\tau = -0.0028/0.0019 = -1.47$; returns: $\tau = -0.92/0.020 = -46.0$
- 2. Price: $-1.47 > -2.86$: do not reject a unit root; returns: $-46.0 < -2.86$: reject
- 3. KPSS: price $2.10 > 0.463$: reject stationarity; returns $0.09 < 0.463$: do not reject
- The price is $I(1)$, the returns $I(0)$: the picture of a random walk, but not yet a proof of it.

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A8: When the Tests Disagree [Proposed]

Solution discussed in class

Task

- Log price of a stock, $T = 1500$, constant and trend: $\hat{\gamma} = -0.0105$, $\mathrm{SE}(\hat{\gamma}) = 0.0029$; KPSS (trend) $= 0.21$. Model: A7.
- 1. Compute τ and decide with the 5% critical value for a constant and a trend.
- 2. Decide with KPSS.
- 3. Combine the two decisions and give two possible reasons for the conflict.
- Report: τ, two decisions and two sentences.

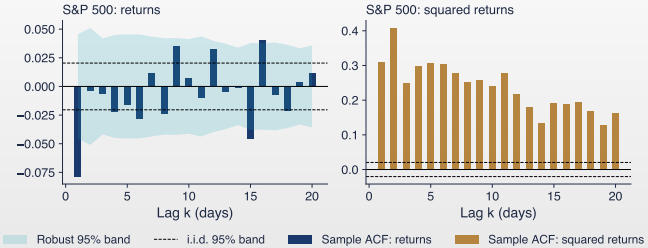
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B1: Autocorrelation in the S&P 500 [Solved]

- **Question**: does the verdict on the autocorrelation of daily S&P 500 returns depend on whether we allow for volatility clustering?
- **Inputs**: S&P 500 closes since 1990, daily log returns in %
- **Tasks**
  1. Draw the ACF of the returns (lags 1–20) with the i.i.d. and the robust 95% bands, and the ACF of the squared returns.
  2. Report $\hat{\rho}(1), \ldots, \hat{\rho}(5)$ of the returns and of the squared returns.
  3. Compute $Q_{LB}(10)$ and the robust $\tilde{Q}(10)$, with p-values, for the returns and for the squared returns.
  4. Say which random walk hypothesis each test rejects.
  5. Interpretation: is weak-form efficiency rejected?
- **Report**: the chart, ten autocorrelations, four statistics with p-values and two sentences
- Notebook: section B1

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B1: Solution [Solved]



- $\hat{\rho}(1), \ldots, \hat{\rho}(5)$, **returns:** -0.079, -0.004, -0.006, -0.022, -0.016; **squared:** 0.311, 0.408, 0.250, 0.298, 0.308; $T = 9\,240$
- **Returns:** $Q_{LB}(10) = 90.8$ (p < 0.001), $\tilde{Q}(10) = 18.8$ (p $= 0.043$); **squared:** $Q_{LB}(10) = 8015$, $\tilde{Q}(10) = 189.4$ (both p < 0.001)
- **Interpretation: squared returns reject RW1 and RW2 (volatility clustering)**
  - **the robust test rejects RW3 only barely, because of a small negative $\hat{\rho}(1)$: a weak, short-lived reversal**

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B2: BET, Bitcoin and Two BVB Stocks [Proposed]

- **Question**: for which of the BET, Bitcoin, Banca Transilvania and OMV Petrom does the verdict on autocorrelation change between the classic and the robust test? Model: B1.
- **Inputs**: BET since 1997, Bitcoin since 2014, TLV and SNP since 2010 (TLV without 30–31 May 2016), daily log returns in %
- **Tasks**
  1. Compute $\hat{\rho}(1)$, $Q_{LB}(10)$ and $\tilde{Q}(10)$ with p-values for each series.
  2. Compute the runs test z and its p-value for each series.
  3. Mark the series for which the classic and the robust test disagree at 5%.
  4. Interpretation: why is the BET different from the other three?
- **Report**: one table and two sentences
- Notebook: section B2

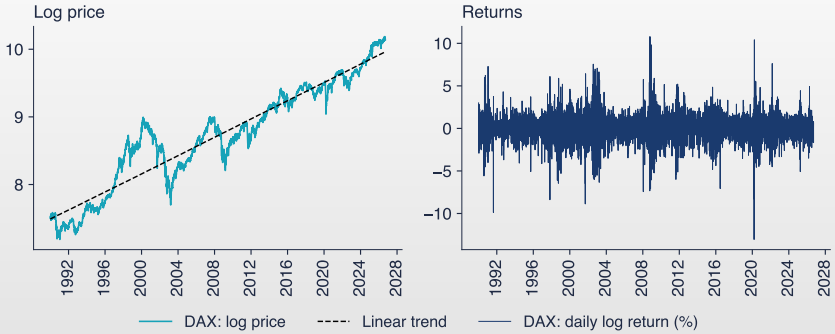
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B3: Unit Roots in the DAX [Solved]

- **Question**: what is the order of integration of the DAX log price and of its returns?
- **Inputs**: DAX closes since 1990; log price and daily log returns
- **Tasks**
  1. Draw the log price with a fitted linear trend, and the returns.
  2. Run ADF (lag order by AIC), PP and KPSS on the log price with a constant and a trend.
  3. Run the same three tests on the returns with a constant.
  4. Interpretation: does a unit root in the DAX price prove that the DAX is weak-form efficient?
- **Report**: the chart, six statistics with p-values and two sentences
- Notebook: section B3

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B3: Solution (1/2) [Solved]



- Left: the DAX log price with a fitted linear trend; right: the daily log returns

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B3: Solution (2/2) [Solved]

	ADF	PP	KPSS
log price (trend)	-2.69 ($p = 0.239$)	-2.65 ($p = 0.260$)	0.94 ($p < 0.01$)
returns (constant)	-23.30 ($p < 0.001$)	-97.40 ($p < 0.001$)	0.04 ($p > 0.10$)

- ADF used 30 lags for the price (AIC); PP and KPSS use 38 Bartlett lags; $T = 9\,288$
- Price: no rejection by ADF or PP, rejection by KPSS: $I(1)$; returns: the opposite: $I(0)$
- Interpretation: no; a unit root is necessary for a random walk but allows predictable increments; efficiency is tested on the returns (B1, B5)

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B4: BET, Bitcoin and TLV [Proposed]

- **Question**: do the three unit-root tests agree for the BET, Bitcoin and Banca Transilvania? Model: B3.
- **Inputs**: BET since 1997, Bitcoin since 2014, TLV since 2010 (without 30–31 May 2016)
- **Tasks**
  1. Run ADF, PP and KPSS on each log price (constant and trend) and on each return series (constant).
  2. Classify each series as $I(1)$, $I(0)$ or inconclusive.
  3. For an inconclusive case, propose one check that could settle it.
  4. Interpretation: would you trade a "mean-reverting" TLV price on this evidence?
- **Report**: one table, three classifications and two sentences
- Notebook: section B4

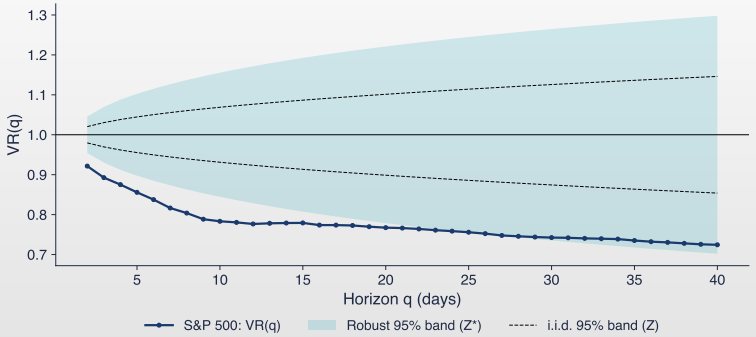
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B5: Variance Ratios of the S&P 500 [Solved]

- **Question**: is the S&P 500 mean-reverting over horizons of 2 to 20 days, once volatility clustering is taken into account?
- **Inputs**: S&P 500 closes since 1990, daily log returns in %
- **Tasks**
  1. Compute VR(q), $Z(q)$ and $Z^*(q)$ for $q = 2, 5, 10, 20$.
  2. Compute the Chow–Denning statistic and compare it with its 5% critical value.
  3. Draw VR(q) for $q = 2, \ldots, 40$ with the i.i.d. and the robust 95% bands.
  4. Interpretation: is the mean reversion large enough to matter for an investor?
- **Report**: a table of twelve numbers, one statistic, the chart and two sentences
- Notebook: section B5

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B5: Solution (1/2) [Solved]



- VR(q) of the S&P 500 for $q = 2, \ldots, 40$, with the i.i.d. and the robust 95% bands

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B5: Solution (2/2) [Solved]

	$q = 2$	$q = 5$	$q = 10$	$q = 20$
VR(q)	0.922	0.856	0.783	0.767
$Z(q)$	−7.54	−6.33	−6.17	−4.50
$Z^*(q)$	−3.37	−2.76	−2.73	−2.06

- $T = 9\,240$; Chow–Denning: $\max|Z^*| = 3.37 > 2.49$ (p = 0.003): reject the random walk
- $Z(2)$ is about 2.2 times $Z^*(2)$: ignoring volatility clustering exaggerates the evidence
- Interpretation: significant, but small
  - the weekly variance is about 86% of five daily variances
  - after trading costs this is hardly a profit

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B6: Five Emerging Markets [Proposed]

- **Question**: which of the WIG20, BUX, PX, BET and BIST 100 departs from a random walk over the last ten years? Model: B5.
- **Inputs**: daily log returns from 19 September 2016 to 18 September 2026
- **Tasks**
  1. Compute VR(5), $Z^*(5)$ and its p-value for each market.
  2. Compute the Chow–Denning p-value ($q = 2, 5, 10, 20$) and the automatic VR test with its wild-bootstrap p-value.
  3. Compute the Šidák critical value for five markets tested together at 5%.
  4. Interpretation: which markets would you call inefficient once you account for testing five markets?
- **Report**: one table, one critical value and two sentences
- Notebook: section B6

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C1: Has the BET Become More Efficient? [Proposed]

- **Question**: how has the predictability of the BET evolved since 1997, including around the FTSE Russell upgrade of September 2020?
- **Inputs**: BET daily log returns since 1997; models: B1, B5
- **Tasks**
  1. Compute VR(5) and $Z^*(5)$ on rolling windows of 500 days, one every 21 days, and the share of windows with $|Z^*(5)| > 1.96$ in the first and in the last five years.
  2. Compute $\hat{\rho}(1)$, VR(5), $Z^*(5)$ and the Chow–Denning p-value in 1997–2006, 2007–2016 and 2017–2026.
  3. Compare VR(5) in the five years before and after 21 September 2020.
  4. List what else changed in the Romanian market at the same time.
  5. Interpretation: how would you separate these changes from the effect of the upgrade?
- **Report**: a table, one chart and a plan for a project
- Notebook: section C1

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C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant whether the DAX (daily, since 1990) is weak-form efficient. The answer:
- (a) ADF on the DAX log price gives tau = −2.69, p = 0.24: the unit root is not rejected, so the DAX is weak-form efficient.
- (b) Ljung-Box Q(10) = 21.83, p = 0.016: DAX returns are predictable, so the DAX is not efficient.
- (c) For squared returns Q(10) = 3747: DAX returns are not a martingale.
- (d) VR(10) = 0.93 < 1 shows positive autocorrelation (momentum) in the DAX.
- (e) Under RW1, VR(q) = 1 at every horizon and the variance of q-day returns is q times the daily variance.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of five verdicts with one line of justification each.

Points to Remember

- A unit root in prices is the starting point, not the answer: efficiency is about the returns
- Test the martingale with robust statistics ($\tilde{Q}$, Z^*): classic tests mistake volatility clustering for predictability
- VR > 1: momentum; VR < 1: mean reversion; the S&P 500 reverts slightly, the old BET trended
- Many horizons or many markets: use Chow–Denning or a Šidák correction
- An AI answer is a draft: check the null of each test, the sign of VR -1 and the robust version

After the Seminar

- Lecture 7 develops each topic of today: efficiency, random walks, ACF tests, unit roots, variance ratios, adaptive markets
- C1 can grow into a team project: BET stocks, WIG20 and BUX as controls, a block bootstrap of the difference
- Reading: Franke, Härdle & Hafner (2019), Ch. 11; exercises in Borak, Härdle & López-Cabrera (2013), Ch. 11; Campbell, Lo & MacKinlay (1997), Ch. 2

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